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Impact of green chemistry education on students' learning and environmental awareness in chemistry

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Abstract
The growing emphasis on sustainability worldwide has made Green chemistry an essential area to study in modern education. The review article examines the impact of Green chemistry education on students' learning outcomes and environmental awareness in chemistry. The article demonstrates how integrating Green chemistry principles into education enhances students' knowledge of chemical processes and develops an enhanced awareness of environmental responsibility through an examination of current studies, educational programs, and pedagogical approaches. Key findings on the importance of Green chemistry education in fostering sustainable development indicate that teaching Green chemistry to students improves their critical thinking skills and academic performance and inspires them to adopt sustainable personal and professional actions. The review also highlights challenges educators face in implementing Green chemistry education, such as limited resources and resistance to curriculum changes, and suggests ways to get beyond them. To provide students with the skills and perspective they need to tackle environmental issues and eventually help create a more sustainable future, the study concludes that Green chemistry education is an effective strategy. The significance of integrating Green chemistry into educational frameworks will encourage the next generation of early-career chemists and researchers about the prospect of Green chemistry and sustainability.

Keywords Green chemistry, Sustainability, Chemistry education, Learning, Environment

1 Introduction

Human activities such as deforestation, industrialization, pollution, and the unsustainable exploitation of resources have increasingly degraded the environment. These actions have triggered major ecological issues, including climate change, the loss of biodiversity, and the buildup of hazardous waste. Tackling these problems necessitates a global approach, with education especially in the sciences playing a critical role in cultivating environmental awareness and promoting sustainability. Chemistry, in particular,

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stands out as a key discipline, as it provides essential knowledge about the composition, behavior, and transformation of matter, which is fundamental to addressing and reducing environmental harm.

Chemistry, often referred to as the central science, is a branch of science that explores the properties, composition, and structure of substances, as well as the changes they undergo during chemical reactions [1]. It is a foundational subject in many scientific disciplines, including biology, physics, agriculture, pharmacy, nursing, biochemistry medicine, and environmental science [1]. Chemistry is fundamental to understanding the physical world and addressing global challenges, including climate change and sustainable energy solutions.

The importance of chemistry extends far beyond the laboratory. It has applications in various industries, ranging from pharmaceuticals to agriculture, and impacts daily life through innovations in materials, food preservation, and waste management [2]. As the world grapples with increasing environmental challenges, chemistry becomes even more significant in developing solutions for pollution reduction, climate change mitigation, and resource conservation. Therefore, chemistry plays a crucial role in addressing global challenges, from combating climate change to developing sustainable energy solutions. However, the traditional approach to teaching chemistry has often overlooked the environmental implications of chemical processes and the need for a more sustainable approach to the field.

Traditional approaches to teaching chemistry, often focused on theoretical knowledge and practical laboratory experiments, have generally neglected to integrate environmental considerations. While these methods effectively teach foundational chemical principles, they tend to overlook the broader ecological and sustainability issues related to chemical practices [3]. The conventional focus on chemical reactions, formulas, and mechanisms may not adequately prepare students to confront the pressing environmental issues of the modern world. Environmental challenges such as pollution, climate change, and resource depletion have intensified over recent decades, prompting a global call for sustainable solutions. The burning of fossil fuels, plastic waste accumulation, and the depletion of natural resources are just a few examples of the growing environmental crises [4]. These challenges require an urgent re-evaluation of how chemistry is taught, with an emphasis on developing sustainable chemical practices and fostering environmental responsibility. As a result, there has been a significant shift toward incorporating environmental considerations into chemistry curricula to help equip students with the tools needed to contribute to a sustainable future [5]. One such approach that aligns with this shift is Green chemistry, an innovative paradigm that emphasizes sustainable chemical practices and environmental consciousness in both teaching and learning.

Green chemistry, which is a part of sustainable chemistry, is a transformative approach within the field of chemistry that focuses on developing environmentally friendly products and processes to minimize waste and prevent the release of toxic substances into the environment [6]. It seeks to minimize the environmental impact of chemical processes by focusing on the design of products and processes that reduce the use and generation of hazardous substances. This field operates under a set of 12 principles that guide chemists toward making more sustainable and environmentally friendly choices in their work as proposed by Anastas and Warner [7], as shown in Fig. 1. These principles not only provide a framework for developing sustainable chemical practices but also serve

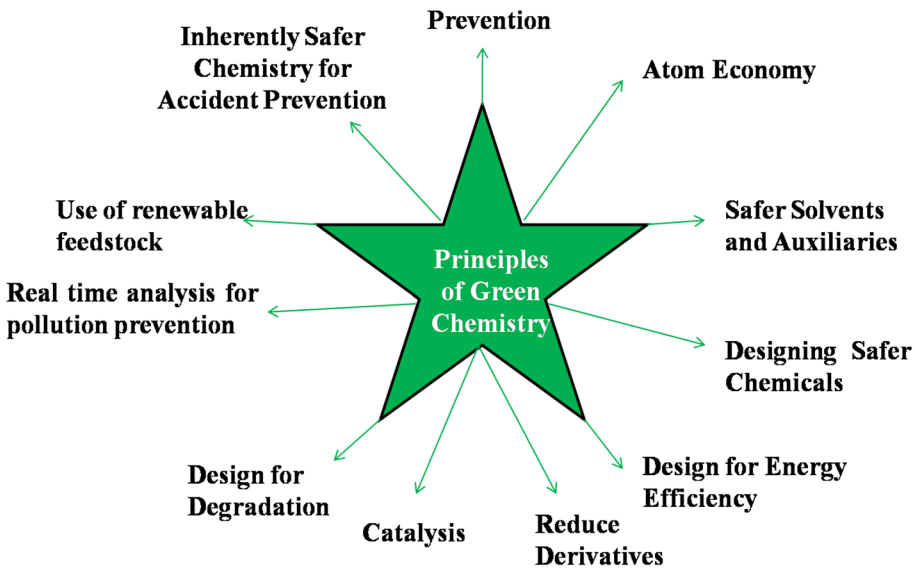

Fig. 1 Principles of Green chemistry [9, 19]

as a pedagogical tool in the classroom, allowing students to learn about chemistry in the context of real-world environmental challenges. By incorporating these principles into the chemistry curriculum, educators can help students develop a deeper understanding of how chemistry intersects with sustainability and environmental conservation [10]. Green chemistry is an important building block for Sustainable Chemistry, but it is not necessarily sustainable, as Green chemistry does not address possible implications of using renewable resources such as total substances, material, product, and energy flows, or alternative business models or catalysis trade-offs if metallic catalysts are used, nor ethics or stakeholder roles, according to a 2021 paper written by a group of sustainable chemistry scholars led by Klaus Kümmerer. Here, Green chemistry is viewed as merely a component of the larger, more formidable structure of sustainable chemistry. "Green chemistry" is only one of ten qualities listed in a list of sustainable chemistry characteristics that was also published in 2021 [7].

The importance of Green chemistry in the classroom is numerous. Not only does it provide students with the knowledge of how chemistry can contribute to solving environmental problems, but it also fosters a deeper engagement with scientific concepts by highlighting the practical applications of chemical principles. Students who are exposed to Green chemistry are more likely to understand the relevance of their studies to global challenges and to develop a sense of responsibility toward the environment [11]. Green chemistry, therefore, can potentially engage students more effectively by connecting theoretical knowledge with practical solutions for sustainability. In addition to enhancing students' understanding of chemistry, Green chemistry is crucial in raising environmental awareness and promoting conservation. Emphasizing the environmental impact of chemical processes, Green chemistry encourages students to think critically about the long-term effects of their actions and decisions. This can lead to increased awareness of sustainability issues, including waste reduction, energy efficiency, and pollution prevention, among students [10]. In addition to developing safety and sustainability standards for chemistry applications, such as environmental justice concepts focused on chemicals management, other organizations have attempted to define and characterize sustainable

chemistry. The Expert Committee on Sustainable Chemistry (ECOSChem), which represents a wide range of constituencies, has built upon these foundations through research, analysis, and in-depth discussion to create a precise and useful definition and set of criteria that can be adopted and modified for use in policy, education, corporate, and investment decision making, among other contexts. Additionally, it is meant to direct the design and application of chemicals, materials, processes, and products in various contexts [14]. Existing research has demonstrated that Green chemistry education can positively influence students' learning outcomes and environmental awareness [12]. Studies have shown that students exposed to Green chemistry curricula tend to develop a better understanding of the chemical principles involved, as well as a heightened sense of environmental responsibility [13].

Furthermore, such educational interventions have been linked to improved attitudes toward sustainability and increased engagement with environmental conservation efforts [15]. However, despite this promising body of research, there remains a gap in the literature regarding the specific impact of Green chemistry on students' academic performance and their broader environmental attitudes in diverse educational settings. This article seeks to address this gap by examining the impact of Green chemistry education on students' learning outcomes and environmental awareness, particularly within the context of secondary school chemistry education. Exploring how Green chemistry influences students' understanding of chemistry concepts and their attitudes toward sustainability, this study aims to contribute valuable insights into the integration of Green chemistry into educational curricula and its potential to promote environmental consciousness among young learners.

1.1 Overview of green chemistry and its principles

For a significant part of its history, chemistry was seen as a science of progress that many people found admirable. But, mostly for environmental concerns, recent public debates on chemistry have been highly critical. Thus, it is notable that "Green chemistry" came into being around the 1990s. A study by Linthorst on Research between Science, Society and Politics: The History and Scientific Development of Green chemistry examine its origins, scientific meaning, and factors that contributed to the remarkable expansion of that emerging discipline. Since its beginning, there has been debate regarding the scientific validity of Green chemistry. With an emphasis on the chemical societies in the USA, UK, and Netherlands, it provides a historical analysis of its origins and evolution. Since Research between Science, Society, and Politics takes into consideration both internal scientific and socio-cultural variables that contributed to the development and expansion of Green chemistry, it should be of interest to chemists, policy makers, and historians and philosophers of chemistry. Linthorst described how Green chemistry originated and developed mainly in the United States, the United Kingdom, and the Netherlands. His research highlights how Green chemistry has been a contentious field since its inception by addressing scientific disagreements and varying interpretations of its scope and meaning. It also places Green chemistry within more general concerns like sustainability, circular economy, and environmental policy, thanks to the contributions of influential people like Paul Anastas, who is known as the father of Green chemistry. Green chemistry is described in Benign by Design: Alternative Synthetic Design for Pollution Prevention (1994) and Green chemistry: Designing Chemistry for the Environment

(1996). He also mentioned that the term "Green chemistry" was rarely used by authors. But! A wide range of chemical knowledge aimed at preventing contamination [16].

Environmental concerns have emerged from the widespread use of hazardous chemicals. A notable example is dichlorodiphenyltrichloroethane (DDT), previously used extensively as an insecticide. Its absorption into soil and subsequent flow into water bodies has created severe environmental impacts, particularly affecting marine ecosystems. Studies have shown that West Bank migrant fish with high DDT concentrations struggle to survive their downstream journey in Portland Harbor [17]. While DDT's environmental impacts highlight the need for safer chemical practices, the concept of Green chemistry by Anastas and Warner [8] provides broader guidelines for the sustainable use and disposal of chemicals. Green chemistry encompasses the development of chemical products and processes that minimize or eliminate hazardous substance usage and production. This field employs ecological approaches to reduce harmful substances in chemical processes and decrease toxic product generation. Green, reactions must incorporate essential components such as environmentally conscious reagents/catalysts, appropriate solvents, and efficient energy consumption. The field also addresses waste management and product degradation planning, aligning with pollution prevention and sustainable development principles. The framework for sustainable chemical design is provided by the 12 principles of Green chemistry. These principles, introduced by Paul Anastas and John Warner [8], serve as comprehensive guidelines for developing new chemical products and processes. They address all aspects of the process lifecycle, from raw material selection to transformation efficiency, safety considerations, and the toxicity and biodegradability of products and reagents. Nowadays, these 12 principles of Green chemistry as shown in Fig. 1 are considered the fundamental to contribute to sustainable development. The principles comprise instructions to implement new chemical products, new synthesis, and new processes [18]. These principles are:

1. Prevention: It is better to prevent waste than to treat or clean up waste after it has been created.
2. Atom Economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. Less Hazardous Chemical Syntheses: Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. Designing Safer Chemicals: Chemical products should be designed to affect their desired function while minimizing their toxicity.
5. Safer Solvents and Auxiliaries: The use of auxiliary substances (solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. Design for Energy Efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/deprotection, and temporary modification of physical/chemical processes) should be

minimized or avoided, if possible, because such steps require additional reagents and can generate waste.

9. Catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. Real-Time Analysis for Pollution Prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. Inherently Safer Chemistry for Accident Prevention: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

1.2 Importance of green chemistry education in fostering sustainable development

Green chemistry education is important in promoting sustainable development as global environmental problems including pollution, resource depletion, and climate change. The importance of Green chemistry education in promoting a sustainable future includes:

1. Promoting environmental awareness: To increase public knowledge of the effects that chemical processes and products have on the environment, Green chemistry education is important. Through the teaching of Green chemistry concepts like waste prevention, the utilization of renewable feedstocks, and efficiency metrics like E-factor (Environmental factor), educators will foster a sense of accountability for reducing environmental damage in their students [20]. Fostering a generation of scientists and people who place a high value on sustainability in both their personal and professional life requires this awareness.
2. Promoting sustainable practices: Adopting and supporting sustainable practices is encouraged when Green chemistry is integrated into the educational setting. For example, students are better equipped to make decisions that lessen their environmental impact when they are taught about safer chemical substitutes and energy-efficient procedures [21]. Beyond the classroom, these activities encourage communities and corporations to adopt more environmentally friendly practices and technologies.
3. Enhancing innovation and problem-solving: Green chemistry challenges students to create solutions that are both efficient and sustainable, which encourages creativity and critical thinking. In addition to improving students' problem-solving abilities, this strategy promotes the creation of sustainable products and innovations [22]. Green chemistry education enables students to address difficult global issues by fostering creativity and transdisciplinary thinking.
4. Supporting global sustainability goals: The Sustainable Development Goals (SDGs), particularly climate change, life below water, and responsible production and consumption, are closely associated with Green chemistry [23]. Curricula that include

Green chemistry give students the skills they need to solve urgent environmental problems.

5. Addressing industry needs: Professionals with expertise in Green chemistry are in greater demand as industries place a greater emphasis on sustainability. According to Hack and Hutchison [24], Green chemistry education addresses this gap by preparing students for jobs in industries including environmental remediation, sustainable manufacturing, and renewable energy. Because education and industry needs are in conformity, graduates will be prepared to promote sustainable growth [25].

2 Learning theories in sustainability and environmental education

Sustainability and environmental education are essential for promoting environmentally friendly behaviors and practices as well as addressing global environmental challenges. The integration of Green chemistry Education (GCE) into the teaching and learning of chemistry requires a strong theoretical foundation to justify its impact on students' learning outcomes and environmental awareness. Several educational theories explain how students learn about sustainability and develop a commitment to environmental responsibility. The following theories provide insight into how GCE fosters cognitive development, critical thinking, collaborative learning, and behavioral transformation among students as shown in Fig. 2.

1. Constructivist learning theory
   The Constructivist Learning Theory, proposed by Jean Piaget [26] proposed that learning is an active process where individuals construct their own knowledge based on experiences and prior understanding. Rather than passively receiving information, students actively engage with new concepts, incorporating them into their existing cognitive framework through exploration, questioning, and problem-solving. In the context of Green chemistry Education, the constructivist approach is crucial because it emphasizes active engagement, inquiry-based learning, and hands-on experiences. Traditional chemistry instruction often relies on memorization and theoretical concepts, but Green chemistry Education encourages students to experiment with sustainable chemical practices, analyze real-world environmental challenges, and develop solutions. For example, instead of merely studying the concept of catalysis, students in a Green chemistry classroom might explore how biodegradable catalysts can replace hazardous ones in chemical reactions. This experiential learning allows them to internalize sustainability principles, making them more likely to apply these concepts beyond the classroom. Moreover, constructivist learning aligns with project-based and problem-solving approaches, which are integral to teaching students about waste reduction, energy efficiency, and green alternatives in chemistry. Engaging

Fig. 2 Learning Theories in Sustainability and Environmental Education

students in discussions and experiments that require them to analyze, evaluate, and apply Green chemistry principles, they develop a deeper understanding of how chemistry contributes to sustainability.

2. Experiential learning theory
   David Kolb's Experiential Learning Theory highlights that learning is most effective when students gain knowledge through direct experiences [50]. The theory presents a four-stage learning cycle:

   - Concrete Experience – Actively engaging in an experience.
   - Reflective Observation – Reflecting on the experience.
   - Abstract Conceptualization – Formulating new ideas based on reflection.
   - Active Experimentation – Applying newly acquired knowledge to real-world situations.

   This iterative cycle ensures that students develop a deeper understanding by continuously applying and refining their knowledge through hands-on engagement. Green chemistry Education is fundamentally experiential because it encourages students to participate in laboratory-based sustainability experiments. When students observe and reflect on the environmental impacts of chemical processes, they conceptualize alternative solutions that are eco-friendly and resource-efficient. For instance, in a Green chemistry lab, students might compare the environmental footprint of traditional solvents versus green solvents in chemical synthesis. Through reflection and experimentation, they understand how their choices affect toxicity, waste generation, and sustainability. Additionally, field-based learning experiences, such as industrial visits to eco-friendly manufacturing plants or community-based waste management projects, help students see the practical applications of Green chemistry. Such experiences bridge theoretical knowledge with real-world environmental issues, fostering a more profound appreciation for sustainable chemistry practices [21].
3. Socio-cultural learning theory
   Lev Vygotsky's Socio-Cultural Learning Theory [31] argues that knowledge is constructed through social interactions, cultural context, and collaborative learning. Learning is enhanced when individuals engage in meaningful discussions, receive mentorship, and participate in guided problem-solving. One of the key concepts of this theory is the Zone of Proximal Development (ZPD), which suggests that learners achieve higher levels of understanding when supported by peers or instructors. Green chemistry Education benefits significantly from collaborative and discussion-based learning, as sustainability challenges often require interdisciplinary and cooperative problem-solving. In Green chemistry classrooms, students work together to evaluate chemical processes, propose sustainable alternatives, and discuss the ethical and environmental implications of industrial practices. For example, a group of students analyzing the carbon footprint of common chemical reactions might collaboratively design low-energy, waste-minimizing alternatives. In this process, they share insights, challenge assumptions, and refine their understanding through peer discussions and instructor guidance. Furthermore, mentorship programs, workshops, and industry-academic collaborations enhance students' learning experiences by connecting them with scientists, policymakers, and environmental advocates who provide real-world perspectives on Green chemistry implementation. The socio-cultural approach also

extends to community-based projects, where students engage with local environmental issues such as pollution control, recycling initiatives, and Green chemistry advocacy thus reinforcing their role as active contributors to sustainability efforts [23].

4. Transformative learning theory
   Jack Mezirow's Transformative Learning Theory explains how individuals undergo deep, meaningful learning experiences that reshape their beliefs, values, and behaviors [32]. According to this theory, learning is transformative when it involves:

   - Critical reflection on existing beliefs.
   - Engagement in discourse that challenges assumptions.
   - Personal and behavioural change based on newly acquired knowledge.

   Green chemistry Education serves as a transformative educational experience, as it challenges students to rethink conventional chemical practices and adopt eco-friendly alternatives. When students are exposed to the negative environmental consequences of unsustainable chemistry, they critically reflect on the need for change and develop a commitment to greener alternatives. For instance, a chemistry student learning about the toxicity of synthetic dyes may initially perceive them as a standard part of industrial processes. However, through transformative learning experiences, they begin to explore biodegradable, plant-based dyes and understand how these alternatives reduce pollution and health hazards. Over time, such shifts in perception lead to long-term behavioural changes, as students become advocates for sustainable chemistry practices. Additionally, transformative learning extends to policy advocacy and industry practices, as students equipped with Green chemistry knowledge can influence industrial sustainability standards, promote ethical decision-making, and contribute to environmental regulations [24].

   The Constructivist, Experiential, Socio-Cultural, and Transformative Learning Theories provide a strong theoretical foundation for understanding how Green chemistry Education (GCE) enhances students' learning and environmental awareness. These theories collectively emphasize active engagement with sustainability concepts, hands-on applications of Green chemistry, collaborative problem-solving, and shifts in perception that lead to long-term behavioural change. Integrating these pedagogical approaches, GCE goes beyond traditional knowledge acquisition, fostering a deep-rooted commitment to environmental sustainability. Through inquiry-based learning, real-world experimentation, and collaborative discussions, students develop the skills and mindset necessary to critically analyze environmental challenges, propose sustainable solutions, and advocate for greener practices. As a result, they move beyond theoretical understanding to become proactive agents of change, influencing sustainability efforts in academia, industry, and society.

2.1 Green chemistry evolution and current trends

The 21 st century has witnessed significant advancements in Green chemistry, driven by growing concerns about environmental sustainability and climate change [25]. Several key trends are transforming the chemical industry include:

Renewable Energy and Feedstocks: Renewable energy sources, such as solar and wind power, are being integrated into chemical processes, reducing dependence on fossil

fuels [26, 27]. Biomass and waste materials are also being utilized as renewable feedstocks, enabling the production of bio-based chemicals. This trend is driven by the need to reduce greenhouse gas emissions and mitigate climate change [8]. Renewable energy sources can power chemical processes, reducing environmental impacts and improving energy efficiency.

Biotechnology and Biocatalysis: Biotechnology and biocatalysis are revolutionizing chemical synthesis, enabling the conversion of biomass into valuable chemicals and fuels [28]. This approach reduces waste generation and minimizes environmental impacts. Biocatalysts, such as enzymes and microorganisms, can be used to convert biomass into chemicals, reducing the need for fossil fuels and minimizing waste generation [29]. Biotechnology can also be used to develop new bio-based materials, such as bioplastics and biofibers [24].

Green Solvents and Separation Technologies: The development of green solvents and separation technologies is crucial for reducing environmental impacts [33]. Alternative solvents, such as ionic liquids and supercritical fluids, are being explored to replace traditional solvents as shown in Fig. 3. Additionally, new separation technologies, such as membrane separation and crystallization, can be used to reduce energy consumption and waste generation [34, 39].

Circular Economy and Waste Reduction: The circular economy concept is gaining traction, emphasizing waste reduction and the reuse and recycling of materials. Chemical companies are adopting circular economy principles to design more sustainable products and processes. This trend is driven by the need to conserve resources, reduce waste, and mitigate environmental impacts. The circular economy approach can help reduce waste generation, promote the reuse and recycling of materials, and improve the sustainability of chemical processes [35].

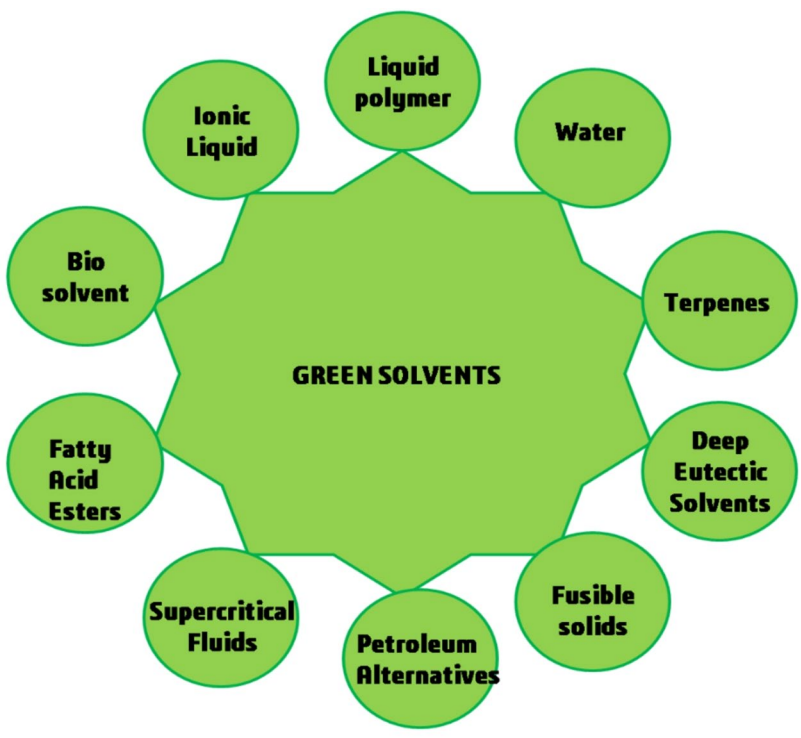

Fig. 3 Different types of Green Solvent

Nanotechnology and Materials Science: Nanotechnology and materials science are enabling the development of new materials with improved properties and reduced environmental impacts [36]. Nanomaterials are being explored for various applications, including energy storage and conversion. These materials can improve the efficiency of chemical reactions, reduce energy consumption, and minimize waste generation. Nanotechnology can also be used to develop new bio-based materials, such as bioplastics and biofibers.

Sustainable Process Design: Sustainable process design is becoming increasingly important, focusing on energy efficiency, renewable feedstocks, and waste minimization [37]– [38]. Life cycle assessment, process simulation, and optimization techniques are being employed to design more sustainable chemical processes. This trend is driven by the need to reduce environmental impacts and improve the sustainability of chemical processes. Sustainable process design can help reduce energy consumption, waste generation, and environmental impacts.

Green Analytical Chemistry: Green analytical chemistry is emerging as a crucial field, focusing on the development of environmentally friendly analytical methods. This approach reduces waste generation and minimizes environmental impacts [40]. Green analytical chemistry can help reduce the risk of environmental pollution and human health impacts. Additionally, this field can promote the development of sustainable analytical methods.

2.2 Level of integration of green chemistry in schools and universities

Green chemistry, as defined by Anastas and Warner [8], is a philosophy that seeks to reduce or eliminate the use or generation of hazardous substances in chemical processes. As the world grapples with environmental challenges such as climate change, pollution, and resource depletion, the importance of Green chemistry education cannot be overstated. Research has shown that professors and academic departments are increasingly integrating Green chemistry concepts into their curriculum and research [41]. A study on Green chemistry education in higher education found that students who participated in Green chemistry programs demonstrated a better understanding of sustainability and Green chemistry principles [42]. Additionally, Green chemistry education was found to positively influence students' attitudes toward chemistry and the chemical industry [43]. Achu [15] study proves that teachers have a positive attitude towards integrating Green chemistry as a part of the curriculum. They prefer getting trained in this arena, as they believe they play a key role in imparting knowledge to children about sustainability practices.

Chemistry educators play a crucial role in imparting knowledge and fostering Green chemistry principles among students [13]. Despite the growing recognition of the importance of Green chemistry education, several barriers to integration have been identified. These include limited resources, lack of teacher training, and inadequate curriculum support [44]. Researchers in the field of education aim to challenge the instrumental rationality linked to Green chemistry, outlining the necessity of establishing a socio-critical viewpoint in the creation and utilization of chemical knowledge, which is known as eco-reflective, humanizing, or building education. Marques and Machado [59] suggest in a study on didactic experiences with Green chemistry that expanding the theoretical underpinnings is essential to taking a critical stance against Traditional Chemistry

practices. Therefore, new pedagogical and philosophical foundations are required for Green chemistry education in order to produce responsible citizens who can participate in the decision-making process regarding chemistry-related environmental challenges. The development of Green chemistry education frameworks is essential for supporting the integration of Green chemistry into education. Teaching methods encompass a wide range of options, including both practical applications and theoretical frameworks. The debate about the forms, tools, and goals of teaching about the subject matter has expanded in scope and complexity since Jan Amos Comenius (1592–1670). A number of writers from various study traditions, including German and Anglo-Saxon, mention the concept of a unique didactic that teaches particular subjects. Kansanen [60] claims that the word didactic has a negative meaning in the Anglo-Saxon culture and is connected to religious indoctrination. The term didactic (Didaktik) in German and Scandinavian traditions, on the other hand, refers to teaching procedures generally, including their technical, moral, and political facets, with the characteristics of an educational science and a manual for teachers' preparation and execution. Since each professional activity has its requirements, methods, and objectives, incorporating GC (as well as its practices and contents) into the education of various professionals would necessitate innovative teaching methods that go beyond antiquated conventional approaches. The core of these traditional teaching methods is the teacher's exposure and transfer of knowledge. The teacher masters logically structured and ordered content and disseminate it to pupils who, in theory, lack knowledge and learn at different rates [61].

To varying degrees, traditional teaching methods may emphasize one or more of such traits. The development of critical thinking, citizenship, and problem-solving skills requires atypical teaching methodologies, despite the fact that rote learning (which emphasizes memory) and traditional approaches in general have a place in education. Based on progressive ideals, many educators have vehemently opposed conventional teaching methods. Paulo Freire argues, for instance, that the disconnection between school curriculum and students' lives diminishes the value of education and reinforces social dominance systems. In order to develop students' formative process, he suggests a dialogical education that considers their active participation, exposing conflicts and attempting to resolve them [62].

2.3 Impact of green chemistry education on students' learning outcomes

Green chemistry approach is an ideal focus for school science education because it presents a modern version of the traditional Chemistry curriculum; uses less toxic materials, making experiments safer for students; teaches critical thinking skills; fewer toxic waste disposal fees; merges scientific concepts with sustainability and responsible human; gives school students opportunities to participate in meaningful science fair and projects. Thus, the use of the Green chemistry approach in the school curriculum has much relevance. Green chemistry approach is a critical component of chemistry Education. It is interdisciplinary in that it provides an opportunity for the sustainable development of the three pillars of a nation (economic, social, and environmental development) [45]. To increase students' awareness on resolving environmental problems, Karpudewan et al. [45] integrated Green chemistry into education programs for both students and teachers.

Teaching Green chemistry in the chemistry classroom is one way of showing the importance of Chemistry for sustainable development [46]. This offers a golden opportunity for students to learn that Chemistry is evolving in such a way that it is becoming more environmentally friendly by focusing on the development of harmless and non-polluting chemicals and chemical processes. Research indicates that Green chemistry education significantly enhances students' learning outcomes by fostering critical thinking and problem-solving skills [47]. Emphasizing the principles of sustainability and environmental responsibility, students are better equipped to connect theoretical knowledge to real-world applications [48]. Therefore, incorporating Green chemistry concepts into curricula will improve students' comprehension of complex scientific ideas by providing practical, relatable examples. Additionally, this approach encourages inquiry-based learning, where students actively participate in experiments that highlight the relevance of chemistry in addressing environmental challenges, thereby increasing engagement and retention. Moreover, Green chemistry education positively impacts students' attitudes toward environmental issues and motivates them to adopt sustainable practices. A quasi-experimental study involving 44 eleventh-grade students showed that those who participated in Green chemistry experiments had significantly higher environmental care attitude scores compared to those in traditional chemistry classes. The experimental group's post-test scores improved markedly, with a p-value of 0.000, indicating strong statistical significance. The p values for the pretest and posttest scores are 0.189 and 0.037, respectively, according to the findings of the independent samples t-test (~α=5%). Furthermore, the paired samples t-test findings showed that the experimental group's p-value was 0.000 and the control group's was 0.349. This indicates that the experimental group's environmental concern attitude score differs significantly from the other groups. Following therapy, the experimental group's scores increased more than those of the control class [52].

In another study by Ijioma [53], 250 participants were chosen using a stratified random sampling procedure from five Nigerian universities out of a total population of 400 undergraduate chemistry students and faculty members. Students' understanding of Green chemistry principles was assessed both before and after the introduction of specific Green chemistry modules through pre- and post-intervention assessments. A pilot research was used to test the study's reliability utilizing a mixed-method approach that included questionnaires and experimental evaluations (Cronbach's alpha = 0.87). The data was analyzed using statistical techniques like descriptive analysis and paired t-tests. With post-intervention test scores rising by an average of 42%, the results show a notable improvement in students' comprehension and application of sustainable chemical processes. Furthermore, 78% of students expressed willingness to adopt Green chemistry in industrial settings. 85% supported making Green chemistry a core part of the curriculum. Research by Santosa et al. [51] highlights that students exposed to Green chemistry principles are more likely to develop pro-environmental behaviours and a sense of responsibility for ecological conservation. This educational framework also promotes interdisciplinary learning, as students explore the intersections of chemistry, environmental science, and social responsibility. Such exposure not only enhances academic performance but also prepares students to tackle global challenges, making them more competitive and conscientious in their future academic and professional pursuits. In North Macedonia in 2024, 211 eighth-grade students participated in a

mixed-methods study that revealed that although the experimental and control groups' academic performance was statistically comparable, students who participated in Green chemistry activities reported increased motivation and involvement and improved conceptual knowledge of chemistry and sustainability [54]. United Nations Environmental Programme (UNEP) emphasizes that Green chemistry education equips learners with skills to address global sustainability issues, including pollution reduction, resource conservation, and ethical chemical design [54]. A study published by the Royal Society of Chemistry highlights that Green chemistry fosters critical thinking, ethical reasoning, and systems-level understanding are skills essential for tackling climate change, resource scarcity, and environmental justice [55].

2.4 Impact of green chemistry education on students' environmental awareness

Environmental awareness, defined as the knowledge, understanding, and sensitivity individuals have about environmental issues, is a crucial component of sustainable development. For students, fostering environmental awareness is critical, as it serves as the foundation for developing sustainable behaviours. Educational interventions have been shown to enhance environmental knowledge, awareness, and pro-environmental actions. Incorporating Green chemistry principles into the secondary school curriculum has been demonstrated to significantly improve students' awareness of waste reduction, pollution prevention, and sustainable resource management [10]. Using a survey research approach, a study shows 475 chemistry students from 12 public secondary schools (SS3) spread across two states in southwest Nigeria. There were twenty-six (26) statements on a four-point scale in a self-structured questionnaire. Cronbach Alpha reliability yielded a rating of 0.965, which indicates strong reliability. The chemistry students' answers were examined through the use of descriptive statistics and correlation analysis to test hypotheses. The results indicated that although their degree of understanding is modest, secondary chemistry students are moderately aware of Green chemistry. They also have a strong impression of how Green chemistry can be applied to actual circumstances. The students' degree of awareness, knowledge, and perception on the use of Green chemistry showed significant correlations [56].

Research demonstrates that Green chemistry education plays a pivotal role in raising students' environmental awareness. Awoyode [10] studies on awareness and application of Green chemistry principles in promoting environmental sustainability among secondary school students in Lagos State showed that secondary school students are aware of the Green chemistry principles. Hence, integrating sustainable practices and eco-friendly principles into the curriculum, students will develop a deeper understanding of the environmental consequences of chemical processes and human activities. It was also revealed that students engaged in Green chemistry lessons are more likely to recognize the importance of reducing waste, minimizing hazardous substances, and utilizing renewable resources. This heightened awareness enables them to critically analyze and propose solutions to environmental problems, fostering a mindset geared toward sustainability. Additionally, Green chemistry education instills a sense of responsibility and stewardship for the environment among students. Through hands-on activities, such as waste recycling projects and green laboratory experiments, learners gain practical insights into how chemistry can be leveraged to protect natural ecosystems. Studies by Santosa et al. [51], show that such experiences not only improve students' knowledge

of environmental issues but also inspire long-term behavioral changes, such as adopting sustainable practices in their daily lives. As a result, Green chemistry education empowers students to become proactive advocates for environmental conservation, contributing to a more sustainable future. Dr. Amy Cannon and John Warner co-founded Beyond Benign, a non-profit organization, in 2007 to empower educators committed to using Green chemistry to bring about a significant, long-lasting, and systemic change in education. In order to assist colleges, universities, schools, and centers throughout the world in integrating Green chemistry into their curricula, Beyond Benign developed the Green chemistry Commitment (GCC) in 2013, with 13 member institutions. This expanding program, which now includes more than 150 institutions across more than 15 countries, aims to promote a varied, approachable, motivating, and empowered Green chemistry community that can bring about the change required for a sustainable future. Students educated under this initiative learn how to design safer chemicals, evaluate environmental and human health risks, and engage in lifecycle assessments for sustainable innovation. This educational framework prepares them to take leadership roles in green industries, policymaking, and advocacy efforts [57].

Also, the Royal Society of Chemistry's 2022 youth report [58] reveals that 81% of young learners believe sustainability and climate education are critical components of their schooling. When Green chemistry is integrated into lesson plans, students report higher engagement and increased confidence in applying science to real-world problems. Such education fosters not only technical proficiency but also climate literacy, allowing students to interpret scientific data, understand environmental policies, and communicate sustainability concepts with clarity and conviction.

2.5 Application of green chemistry in laboratory practices

Adding to the existing efforts, the application of Green chemistry in laboratory module's primary objective is to introduce the next generation of environmentally conscious engineers and chemists to both the green and non-green elements of the synthesis of noble metal nanoparticles (NPs). The suggested lab module adds another element of innovation by integrating guided, inquiry-based learning, which is different from the earlier nanolaboratory educational work. In summary, a similar research problem was presented to undergraduate and graduate students from a variety of fields, including chemistry, physics, materials engineering, and sustainability. They were free to develop an informed answer of their own. Students in an Experimental Nanomaterials and Nanoscience laboratory course created AgNPs and AuNPs using the nongreen Lee–Meisel and Turkevich procedures in the first lab activity. Using the nongreen Lee–Meisel and Turkevich procedures, students in an Experimental Nanomaterials and Nanoscience laboratory course created AgNPs and AuNPs in the first lab activity segment. Based on their degree of expertise and understanding, students were asked to choose and use a green fabrication technique in the second lab activity. A new methodology that promotes teachers and students as learning partners is inquiry-based learning. In actual scientific situations, a researcher must apply his cognitive skills and scientific reasoning to solve the experimental problems when confronted with several open issues. To become an expert in the technical sector, one may need more than just the knowledge that is typically given in a classroom. This teaching methodology has been successfully included into the undergraduate curriculum and a number of primary and upper-grade schools

throughout the last ten years. Hence, in contrast to traditional laboratory experiments, learning through guided scientific inquiry is anticipated to enhance students' interest in nanochemistry and expand their comprehension and learning capacities [64].

Students used the Lee–Meisel [65] and Turkevich [66]– [67] methodologies to synthesize colloidal AgNPs and AuNPs at the test tube level in the first lab activity segmen $t._{3l4}.3_2aC_{657}.2_2-]_{3l4}.3_2$ Following the conclusion of the laboratory, the teaching assistants verified the reproducibility of the aforementioned processes [68]– [71]. The National Research Council (NRC) established five key components for inquiry-based teaching and learning, which served as the basis for the development of this lab module. Using this method, students: "Are interested in scientifically oriented questions, Prioritize evidence, which enables them to formulate and assess explanations that address scientifically oriented questions, Evaluate their explanations in light of alternative explanations, especially those that reflect scientific understanding, and Communicate and justify their proposed explanations [64].

Another example is the Emergy analysis, "EMbodied enERGY" is the root word "emergy." Emergy analysis is a method that relates to the conceptual framework of systems thinking. Actually, in emergy diagramming, a system's states are represented by tuples of numbers, each of which corresponds to a price of stock. The hierarchical level of the intended description and the overall goal of the study determine which collection of variables should be used. The stocks are the names given to these variables in the systems language. In addition to representing actual flows of anything, such as mass, energy, or information in the case of emergency analysis, the links between stocks are also logical, as in network analysis. Insofar as they impact the system's dynamics, the feedback is always explained explicitly. Interesting examples are provided in a variety of fields of study, such as those where the integrated evaluation of an industrial process's sustainability requires the evaluation of the energy carried by the chemical generated or utilized in the system, which in turn demands the evaluation of the emergy [79].

2.6 Importance of systems thinking in green chemistry

Systems thinking help chemistry teachers see how chemistry relates to the dynamic, complex social, technological, economic, and environmental systems that govern our world more globally than reductionist methods that offer fragmented knowledge of chemical reactions and processes. In order to visualize the relationships and connections between the various components of a system, analyze how the system's behaviour evolves over time, and comprehend how interactions between the system's components lead to systems-level phenomena, the new field of systems thinking in chemistry education (STICE) employs tools, techniques, and cognitive frameworks. Formulated by an international IUPAC effort, the STICE framework aims to articulate methods and examples for integrating systems thinking into general chemistry instruction. With three interconnected nodes or subsystems, the STICE framework centers the learner inside a learning system (Fig. 4). Learning theories, learning progressions, cognitive and affective components of learning, and social contexts for learning are some of the processes that are investigated and described by the Educational Research & Theories node for students. Applying broad knowledge of how students learn to the particular difficulties of studying chemistry is what the Chemistry Teaching & studying node does. Understanding how students learn in general is applied to the particular difficulties of learning

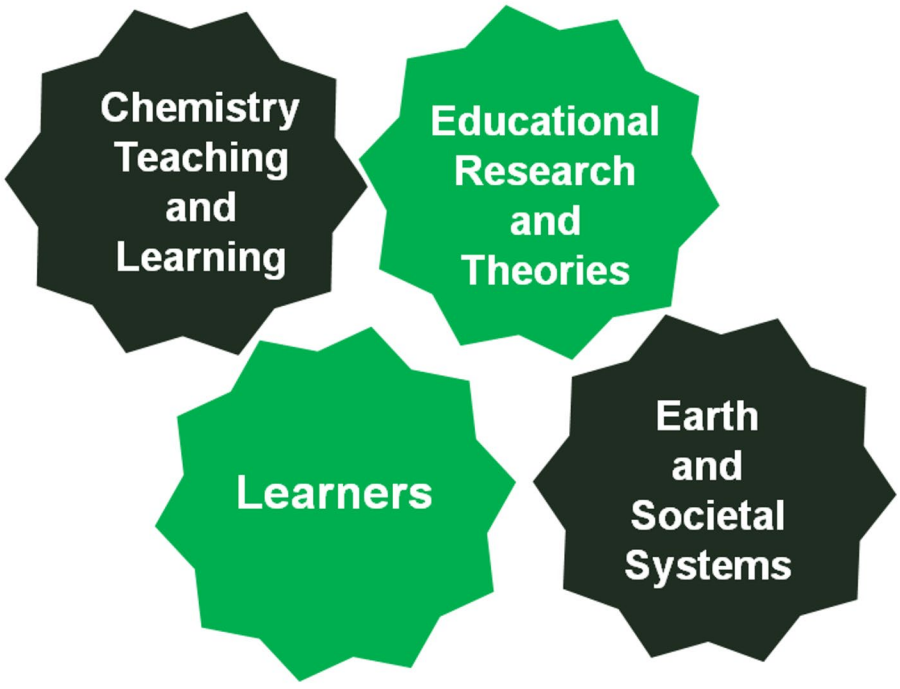

Fig. 4 Systems Thinking in Chemistry Education framework [72]

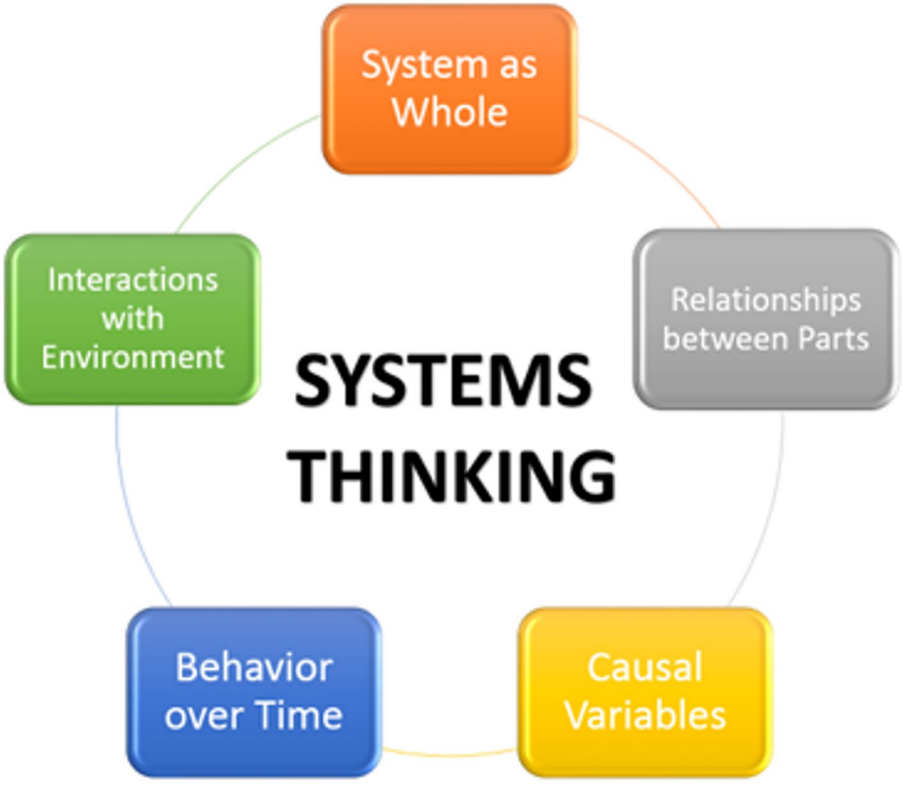

Fig. 5 Intersection between system thinking [78]

chemistry via the Chemistry Teaching & Learning node. Utilizing pedagogical content knowledge, analyzing how the intended curriculum is implemented, evaluated, taught, and applied, and incorporating student learning outcomes that involve accountability for the sustainable and safe use of chemistry are some examples. The Earth & Societal Systems node guides chemistry instruction to address environmental and societal demands expressed in programs like the Planetary Boundaries framework and the UN Sustainable Development Goals [72].

The stratospheric ozone earth system, one of the nine earth system processes in this framework, is an example of how the chemical profession has successfully applied systems thinking to stop the movement of a control variable beyond a planetary limit. The 1987 signing of the Montreal Protocol on Substances that Deplete the Ozone Layer [73] followed the discovery that sunlight photolyzes atmospheric CFCs, releasing chlorine atoms that function as catalysts to destroy stratospheric ozone [74]. "It is necessary to bridge traditional scientific disciplines and examine the earth as an interrelated system of physical, chemical, and biological processes," said Richard Benedick, who led the United States delegation in the ozone negotiations [75]. This application of systems thinking resulted in international action that restored the stratospheric ozone depletion control variable below the planetary boundary, setting an example for addressing other borders. Nevertheless, thirty years after the signing of the Montreal Protocol, the limitations of the use of systems thinking are clearly evident. The release of chlorine atoms into the stratosphere has been considerably reduced by short-term (HCFCs) and longer-term (HFCs) replacement compounds for CFCs; nevertheless, the full impact of these replacement compounds on global warming was not completely taken into account when they were implemented. Many of the HFCs have global warming potentials hundreds or thousands of times greater than those of CO_2, but they have zero ozone destruction potential. Chemistry education and practice can be more fully focused toward sustainability by using systems thinking to direct efforts to educate about the molecular basis of sustainability. According to recent research, systems thinking might be helpful to enhance comprehension of difficult environmental issues like climate change. In order to better address the sustainability of the earth and societal systems, we analyze general chemistry coverage of the Haber-Bosch process and provide examples of current practice and methods that systems thinking could provide such reorientation [72].

$$\mathrm{N_2\,(g)} + 3\mathrm{H_2\,(g)} \rightarrow 2\mathrm{NH_3}\,\text{(In the presence of catalyst and high temperature and pressure)}$$

The framework and resources offered by systems thinking enable general chemistry teachers to investigate how to relate this chemical reaction to nitrogenous compound planetary cycles and to integrate the study of fundamental chemistry ideas into conversations about sustainability agendas like the UNSDGs and the planetary boundaries framework. Students are introduced to the use of systems thinking and are given a clear illustration of the significance of the molecular foundation of sustainability by extending the scope of the reaction's typical presentation. While atmospheric nitrogen, or N_2, makes up 78% of the gaseous component of air, most organisms cannot directly use this atmospherically abundant species to form the nitrogenous compounds that make nitrogen the fourth most abundant element of cellular biomass because it is too unreactive [76]. A century and a half ago, it was acknowledged that the amount of reactive or fixed

Table 1 Distinctive features between systems thinking approach and Context-Based learning approach [78]

Systems thinking approach	Context-based learning approach
Investigates the linkages and connections that exist between a system's components and how they cause cyclic system behaviours. (Fig. 5)	Encourages active student-teacher and student-student engagement in a community of practice.
Recognizes a system not only as a group of components but as a whole.	Positions a learning exercise in a relevant, practical setting.
Determines the factors that influence system behaviour, including distinctive emergent behaviours at the system level.	Discusses important concepts in chemistry and requires that students complete assignments that are related to chemistry.
Investigates how system behaviours evolve over time.	uses language unique to chemistry to engage students
Identifies how a system interacts with its surroundings, including the environment's human components.	Enhances students' existing knowledge

nitrogen available limits plant growth. In 1898, Sir William Crooks called on chemists to "come to the rescue" of endangered European communities and use their "genius" to find ways to produce reactive and bioavailable nitrogenous compounds on a global and industrial scale. He noted that new sources of fixed nitrogen were needed to replace Chilean saltpeter in order to prevent mass starvation [77]. For students who might not have had much experience analyzing and synthesizing system components or seeing dynamic interconnections between isolated chemistry content and the flow of material and energy through earth and societal systems, using a systems thinking approach to broaden the scope of reductionist coverage of the Haber-Bosch reaction to a broader perspective presents both opportunities and significant challenges.

2.6.1 Relationship between systems thinking and other methods to chemistry education reform

Scientists and researchers have raised concerns about conventional approaches to teaching chemistry, claiming that they fail to inspire students, promote meaningful learning, or equip them to use their knowledge of the subject in practical settings. Chemistry education, for instance, has been characterized as fragmented, disconnected, superficial, algorithmic, and decontextualized in recent writings. Numerous attempts have been made to resolve these objections and enhance the results of chemistry instruction. There are two major categories of efforts that have been used to change chemistry education, some of which we highlight here, even though we acknowledge that it is especially difficult to fully separate subject from presentation. Reforming the pedagogies used in chemistry courses, such as POGIL (Process Oriented Guided Inquiry Learning), PLTL (Peer-Led Team Learning), context-based learning, problem-based learning (PBL), case-based learning (CBL), and argument-driven inquiry (ADI), has been the main goal of the first set of approaches. Other reform initiatives have mostly concentrated on the material covered in chemistry classes, defining what is necessary for students to understand and then categorizing that information into certain themes or categories that are then reviewed over the course of a course or study program. Chemical Thinking, Chemistry, Life, the Universe, & Everything (CLUE), Chemistry Unbound, and Chemistry are a few examples of this kind of approach to chemistry education reform. Another approach used to address some of the criticisms about chemistry teaching is the use of systems thinking as shown in Table 1 [78].

2.7 Challenges faced by educators in implementing green chemistry education

Implementing Green chemistry education presents several challenges for educators. A significant obstacle is the lack of adequate resources and infrastructure to support sustainable practices in teaching laboratories. Additionally, the scarcity of standardized Green chemistry curricula poses difficulties in uniformly integrating these principles across diverse educational settings. This lack of standardized curricula can lead to inconsistencies in the depth and breadth of Green chemistry education provided to students. Another challenge lies in the limited awareness and understanding of Green chemistry principles among educators themselves [1]. Many instructors have been trained in traditional chemical methodologies and may not be familiar with the latest advancements in Green chemistry. This gap in knowledge can hinder the effective teaching of Green chemistry concepts, as educators may feel unprepared to incorporate new materials into their courses. Furthermore, the absence of professional development opportunities focused on Green chemistry exacerbates this issue, leaving educators without the necessary support to transition to more sustainable teaching practices [13]. Moreover, institutional resistance to change can impede the adoption of Green chemistry education. Educational institutions often have established curricula and teaching methods, and introducing new content requires overcoming bureaucratic hurdles and resistance from stakeholders accustomed to traditional approaches. This resistance can be due to a variety of factors, including concerns about the efficacy of new teaching methods, the perceived relevance of Green chemistry, or the potential disruption to existing programs. Additionally, the interdisciplinary nature of Green chemistry necessitates collaboration across different departments, which can be challenging in institutions with rigid departmental boundaries. Addressing these challenges requires concerted efforts to raise awareness, provide adequate training for educators, develop flexible curricula that can adapt to the evolving landscape of chemical education [49] and also apply REACH. REACH aims to regulate chemicals that were not previously covered by other laws. This regulation, which goes by the acronym, establishes new requirements and processes for registering (R), evaluating (E), and authorization (A) chemical substances (CH) or limiting or outright prohibiting their manufacture. REACH is an EU law that was implemented to increase the competitiveness of the EU chemicals industry while protecting human health and the environment from potential chemical hazards. Additionally, in an effort to minimize the amount of animal studies, it encourages the use of alternate techniques for evaluating the hazard of substances. Particular consideration is given to POPs and chemicals that are categorized as carcinogenic, mutagenic, or reprotoxic (CMR). The principle of REACH, states that some hazardous compounds known as "Substances of Very High Concern" (SVHC) must be phased out if safer substitutes provide. Prior to REACH, chemicals could only be prohibited if they were shown to be harmful; now, EU industry and importers must demonstrate that every material meant for the market is safe for the environment and human health. The phrase "reversal of the burden of proof" is occasionally used to describe this. These goals are part of the many steps in the process. Registration applies to chemicals (substances and products) that are meant for the market, as long as the manufacturer and amount of production exceed one ton annually. Numerous particular products are not REACH-registered. Chemicals used in cosmetics, biocide, and agricultural, for example, are not covered by REACH since they fall under another law. Substances used exclusively for research and development, as

well as quantities below the threshold, are excluded from the registration requirements. The application of precautionary principle is explicitly asserted in REACH's Article 1(3). Based on the idea that producers, importers, and downstream consumers should make sure that the substances they produce, put on the market, or use don't have a negative impact on the environment or human health, this regulation was created. The precautionary principle serves as the foundation for its regulations [63].

3 Conclusion

Green chemistry is a transformative approach in the field of chemistry that focuses on developing environmentally friendly products and processes to minimize waste and prevent toxic substances from being released into the environment. The twelve principles of Green chemistry guide chemists towards making more sustainable choices in their work, including prevention, atom economy, less hazardous chemical syntheses, designing safer chemicals, safer solvents and auxiliaries, energy efficiency, renewable feedstock, reduction of derivatives, catalysis, design for degradation, real-time analysis for pollution prevention, and inherently safer chemistry for accident prevention. These principles not only provide a framework for developing sustainable chemical practices but also serve as a pedagogical tool in the classroom, allowing students to learn about chemistry in the context of real-world environmental challenges. Green chemistry education significantly impacts students' environmental awareness, promoting sustainable behaviors and waste reduction. Studies show that secondary school students in Lagos State are aware of Green chemistry principles, leading to a deeper understanding of environmental consequences. Green chemistry education instills responsibility and stewardship for the environment through hands-on activities like waste recycling and green laboratory experiments. This education empowers students to become proactive advocates for environmental conservation, contributing to a more sustainable future. However, implementing Green chemistry education presents challenges for educators. The lack of adequate resources and infrastructure to support sustainable practices in teaching laboratories, the scarcity of standardized curricula, and limited awareness among educators can hinder effective teaching of Green chemistry concepts. Additionally, institutional resistance to change and collaboration across different departments can impede the adoption of Green chemistry education. Future studies should examine various geographic and cultural contexts, as well as primary, secondary, and tertiary educational settings. To evaluate the long-term impacts of Green chemistry education on students' environmental behaviors and job choices, longitudinal studies are also required. The effectiveness of the Green chemistry curriculum should be assessed using standardized frameworks. In addition to academic achievement, these instruments ought to track shifts in students' environmental attitudes, awareness, and behaviors.

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Author contributions
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